Eventstudy Analysis for Displacement Events

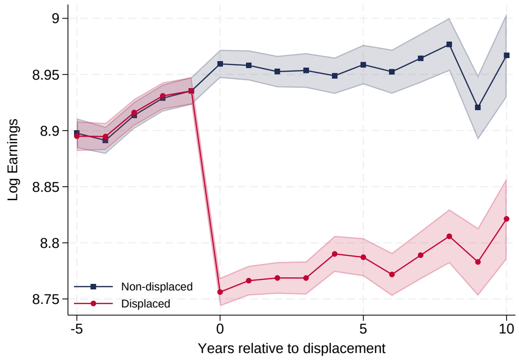
1 Comparing Eventstudy Specifications

Next, we compare the eventstudy estimates across different specifications.

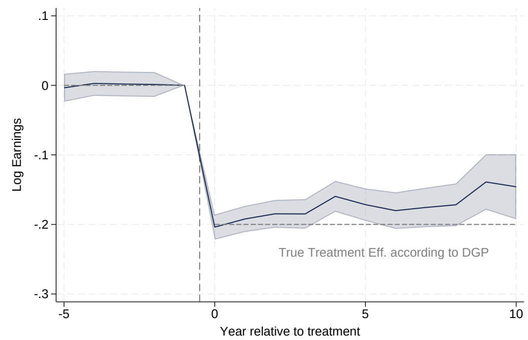
- Raw Means (always good to compare to the eventstudy estimates)
- OLS - Rel. year effects, no person FE
- JLS Specification: Cal. year and person FE - since we are not cotrolling for relative year effects, the estimates are not consistent
- Rel. year effects, person FE
- Relative year, calendar year and person FE
- Controlling for fully interacted calendar and relative year effects, as well as person FE

The matching design does the heavy lifting, that's why these specification yield very similar results.

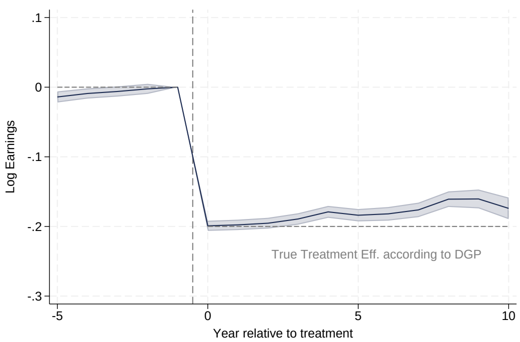
Figure 1: Comparing Eventstudy Specifications



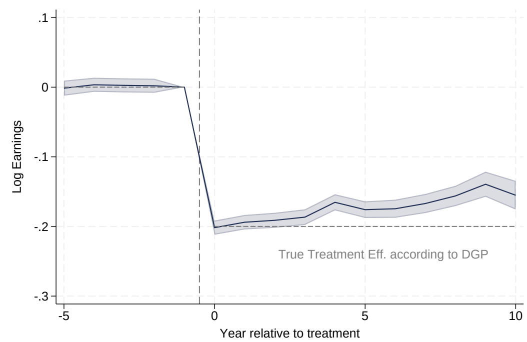
(a) Raw Means



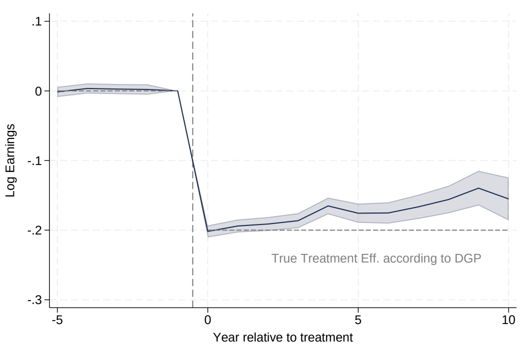
(b) Rel. year effects, no person FE



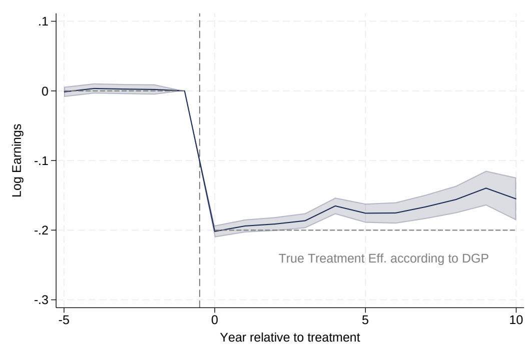
(c) Cal. year and person FE - does not work!



(d) Rel. year effects, person FE



(e) Rel. year, cal. year and person FE



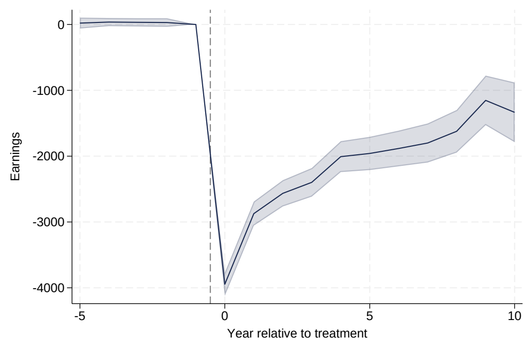
(f) Rel. year $\times$ cal. year and person FE

2 Different Outcomes

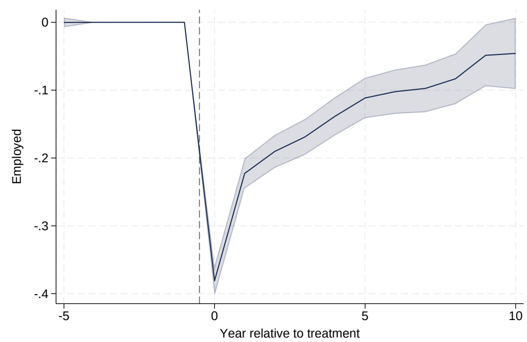
Next, we create eventstudy estimates for different outcomes.

- Earnings in Levels
- Earnings Growth
- Log Earnings conditional on positive earnings
- Employment

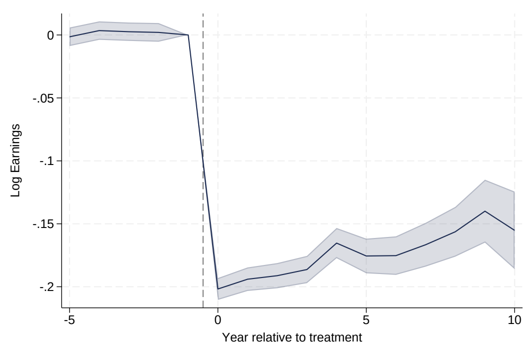
Figure 2: Event Study Estimates for Different Outcomes



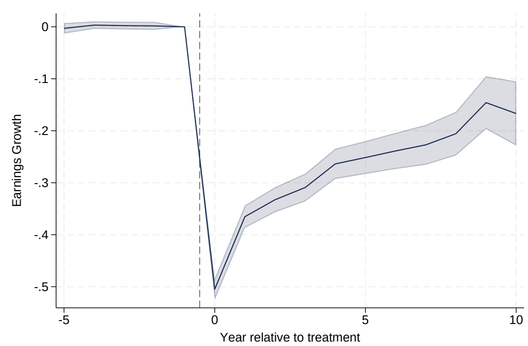
(a) Earnings in Levels



(b) Employment



(c) Log Earnings cond. on pos. Earnings



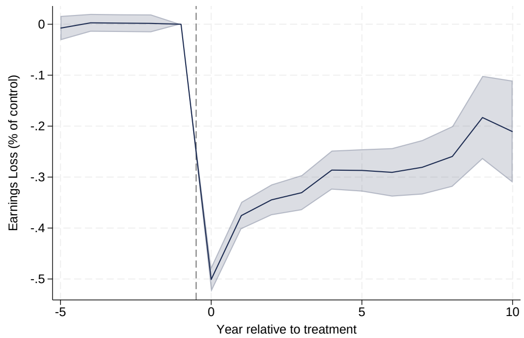
(d) Earnings Growth

3 Different Outcomes

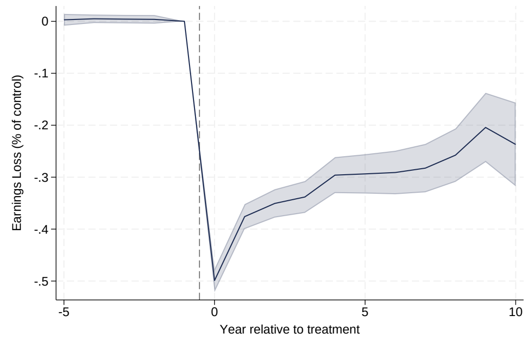
Next, we create eventstudy estimates for different outcomes.

- Scale Raw Mean Difference by Control Mean:
  - Outcome: $\frac{\bar{y}_t^D - \bar{y}_t^{ND}}{\bar{y}_t^{ND}}$
- Scale Regression Coefficient from level regression by Control Mean:
  - Outcome: y_{it}
  - Regression: $y_{it} = \alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k + \varepsilon_{it}$
  - Scale: $\frac{\hat{\beta}_t}{E[y_{it}|X_{it}, Non-Displaced]}$
- Earnings Growth in percent:
  - Outcome: $\Delta y_{i,t} = \frac{y_{i,t} - y_{i,-1}}{y_{i,-1}}$
  - Regression: $\Delta y_{i,t} = \alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k + \varepsilon_{it}$
  - Scale: β_t
- Poisson QMLE:
  - Outcome: it
  - Regression model: $E[y_{it}|X_{it}] = \exp(\alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k)$,
  - Scale: $\exp(\hat{\beta}_t) - 1$

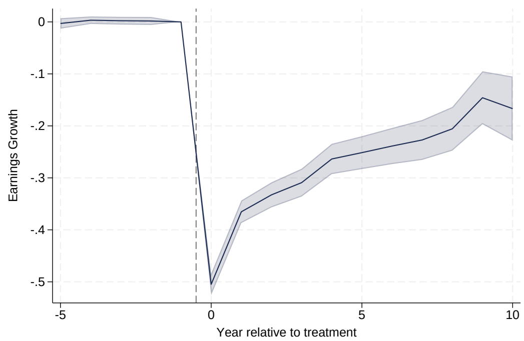
Figure 3: Expressing Earnings Losses in Percent (incl. zero earnings)



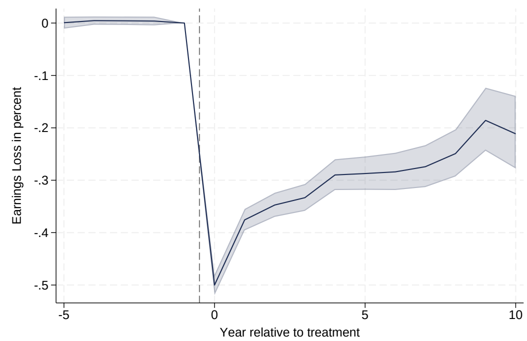
(a) Scale Raw Mean Difference: $\frac{\bar{y}_t^D - \bar{y}_t^{ND}}{\bar{y}_t^{ND}}$



(b) Scale Regression Coefficient: $\frac{\hat{\beta}_t}{\bar{y}_t^{ND}}$



(c) Earnings Growth: $\frac{y_{i,t} - y_{i,-1}}{y_{i,-1}}$



(d) Poisson QMLE, scaled to %: $\exp(\hat{\beta}_t) - 1$